FluencyScore: A Structured LLM-Judge for Fluency in Machine Translation

Airana Mongush[1]
Algebras AI
aira@algebras.ai

Dmitrii Pukhov
Algebras AI
dima@algebras.ai

Abstract

LLM-based machine translation evaluation has advanced rapidly since (Kocmi and Federmann, 2023a) showed that zero-shot prompting of GPT models reaches state-of-the-art system-level agreement with MQM ratings, and that the capability emerges only from GPT-3.5 upward. Later work extended the approach in several directions: MQM-style error annotation with PaLM-2, with quality dependent on model scale (AutoMQM; Fernandes et al., 2023); few-shot GPT-4 prompting, ranked among the strongest submissions at WMT23 (GEMBA-MQM; Kocmi and Federmann, 2023b; Freitag et al., 2023); reasoning-based prompting (Lu et al., 2024); and repeated sampling, with GEMBA-MQM V2 aggregating ten GPT-4.1-mini judgments per segment to rank first on the WMT24 MQM test sets (Junczys-Dowmunt, 2025). The official WMT25 metrics likewise run on a GPT-4.1 backbone.

At the same time, several results indicate that further correlation gains along this axis are limited. Judge accuracy does not consistently track model capability (Tan et al., 2025); human preference data measurably favors fluent but incorrect responses (Sharma et al., 2024; OpenAI, 2025); and the design of the scoring scheme itself shapes model behavior (Kalai et al., 2025). Part of the headroom above current metrics is therefore annotation noise rather than unmodeled quality.

FluencyScore agrees with expert rankings on 72% of segment pairs on WMT25 ESA and on 78% on WMT22 MQM, ahead of every released baseline on identical segments (MetricX-24-Hybrid, XCOMET, CometKiwi, chrF++; all $p<0.001$), and its pooled segment-level Spearman of 0.61–0.62 is at or above published inter-annotator agreement for these protocols (0.41–0.58; Freitag et al., 2021). To attribute these numbers correctly, we compare against the strongest available baseline: GEMBA-ESA, the family used for the official WMT25 scores, re-run on the same Gemini backbone, the same segments, and the same fair mask. On this matched setup the two frameworks are statistically equivalent in correlation ($\Delta\rho = +0.007$, CI covering zero); of the $+0.065$ gap over the official leaderboard score, $+0.058$ is contributed by the backbone. Variation across language pairs is roughly twenty times larger than any effect of prompt design.

Our hypotheses for the structured framework therefore concern its outputs rather than its score, and the largest confirmed return is diagnostic. For every segment, FluencyScore produces a complete error record: an issue type from a fixed taxonomy, a severity level, the exact problem phrase, and a suggested fix. In addition, the structured variant is substantially more stable under a change of judge, varying within a 0.022 Spearman range across four backbones against 0.128 for a flat single-score variant of the same rubric, and its per-pair behavior is predictable from typological distance. Further framework gains (calibration, sub-dimension weighting, language-pair routing) remain hypotheses under active test.

The relevant question for LLM-based evaluation is thus shifting: well-prompted judges already match human judgment up to annotation noise, so the value of an evaluation lies in what it returns at that level of accuracy, and in whether those returns persist when the underlying model changes. FluencyScore is designed for this setting.

1 Introduction

This section motivates treating fluency as a distinct evaluation axis and states the contributions of the paper.

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1.1 Problem and motivation

Automatic MT metrics such as COMET, GEMBA, and chrF conflate several quality dimensions and do not isolate fluency as a distinct axis. Fluency covers idiomatic phrasing, collocational fit, discourse coherence, pragmatic appropriateness, and freedom from calque; a single scalar rarely exposes where a translation fails on these grounds.

WMT25 ESA and historical WMT MQM benchmarks provide independent human judgments at the segment level (Kocmi et al., 2024, 2025). We use a fair comparison protocol (same segments, no imputation, bootstrap CIs) and complement aggregate correlation with per-segment error analysis.

1.2 Contributions

- A structured, rubric-based LLM-judge (FluencyScore) with five weighted fluency sub-dimensions and a double-pass scoring procedure, plus a monotone calibration (FluencyScore-cal) for reporting.
- A fair-comparison evaluation against official WMT25 metrics and against GEMBA-MQM / FluencyScore-MQM on WMT22–24, with bootstrap 95% CIs and paired bootstrap tests; matched-backbone contrasts are primary.
- A backbone ablation and a typological analysis linking syntactic distance to FluencyScore's advantage.
- A segment-level error analysis with an issue-type taxonomy, headline failure modes, and a variance-collapse study of the MQM-style variant.
- Public numeric artifacts and scrubbed analysis notebooks at `https://github.com/algebras-ai/fluencyscore-wmt-artifacts` (judge prompt and sub-dimension weights remain proprietary; see Appendix A).

2 FluencyScore and FluencyScore-cal

Structured FluencyScore: the judge fills five sub-dimension scores and an error record; we combine the five scores into one scalar aggregate.

Flat variant: the same backbone is asked for a single overall fluency score with no rubric and no error record (our FluencyScore-MQM-flat condition).

FluencyScore-MQM: the same rubric as structured FluencyScore, but scored on a negative MQM-style scale (start high, subtract severity penalties).

FluencyScore scores five fluency sub-dimensions—idiomatic, collocational, discourse, pragmatic, and calque—and aggregates them with fixed weights into a single value on a 1–10 scale. The sub-dimension weights and the full judge prompt are proprietary (Section 11; Appendix B).

FluencyScore-cal is a fixed piecewise-linear, non-decreasing map of the raw aggregate onto a reporting scale that is easier to compare with human ESA/MQM numbers. Because the map is non-decreasing, rank correlations are unchanged up to ties. Parameters were fixed during development and are not fitted on any evaluation cohort in this paper.

2.1 Output schema, taxonomy, and scoring protocol

Reference-free by design. The judge sees the source and the translation only—no reference (product use case: online QE / WMT AutoEval Task 3). A reference-conditioned ablation remains future work.

For each segment the judge returns: (i) five sub-dimension scores on a 1–10 scale; (ii) the scalar aggregate (fixed weighted sum); (iii) one main `issue_type` from {`calque`, `register_mismatch`, `false_friend`, `idiom_literal`, `collocation`, `discourse_marker`, `none`}; (iv) severity; (v) problem phrase; (vi) suggested fix; (vii) confidence fields (`low_confidence`, `max_score_diff`).

Double-pass. The judge scores each segment twice independently; we average the two aggregates. A single-pass control shows the second call adds nothing measurable ($\Delta\rho$=+0.007, CI $[-0.011, +0.026]$).

Discourse contributes near-zero to rank correlation on our masks; we keep it as a diagnostic channel. Weights stay fixed.

3 Experimental setup

Section 3.1 lists experiments; Section 3.2 defines masks and statistics; Section 3.3 names judges; Section 3.4 states reproduction gates.

3.1 Overview of experiments

Table 1 maps each experiment to its data, sample size, and the question it is meant to answer. Exp. 2 and Exp. 3 stay separate so year-fixed depth and cross-year scope each keep their own takeaway.

Table 1: Experiment map. Exp. 2 = single-year depth; Exp. 3 = cross-year scope.

#	Name	Data	n	Main question
1	WMT25 ESA	WMT25, 12 pairs	3,526	vs official WMT25 metrics
2	WMT22 MQM	WMT22, 3 pairs	808	scale/structure at fixed year
3	WMT22–24 MQM	9 year×pair cells	2,628	cross-year stability
4	Framework × backbone	WMT22 + WMT25	808 / 3,526	judge vs framework

3.2 Fair-comparison protocol

Language pairs (WMT25 fair subset). EN→ar_EG, bho_IN, cs_CZ, et_EE, is_IS, it_IT, ja_JP, mas_KE, ru_RU, sr_Cyrl_RS, uk_UA, zh_CN (12 pairs; per-pair n in Table 6).

WMT25. We join official automatic scores with FluencyScore on aligned segments. Starting from 4,560 scored rows, the primary fair mask keeps a row only if human ESA and all of {FluencyScore, GEMBA-ESA, MetricX-24, XCOMET, CometKiwi} are non-null → n=3,526 (retention 66.3%–90.0% by pair). chrF++ is not part of this mask (missing for EN→it_IT); we report chrF++ on its own subset (n=4,200).

MQM. Join on (`year`, `pair`, `seg_id`, `system_id`); keep segments where every compared metric is non-null.

Surface baseline on WMT22–24 (Exp. 3). We recompute chrF++ with sacrebleu (`CHRF`, `word_order=2`) using `mt-metrics-eval-v2 refA` aligned by exact source-text match. Result: Spearman ρ=0.290, PA=0.626, n=2,553 (83 fair rows lack an exact source match and are omitted from the chrF++ column only). This replaces the previous n=818 / WMT22-keys-only figure (ρ=0.351, PA=0.653).

Agreement statistics. Spearman ρ (primary), Kendall τ, and pairwise accuracy PA=$(\tau+1)/2$. Headline ρ/τ/PA are pooled unless a table says otherwise. Table 6 also reports per-pair ρ and a macro-average.

Uncertainty. Bootstrap 95% intervals by segment resampling; paired bootstrap for metric differences.

3.3 Judges per experiment

Table 2 lists the judge backbones used in each experiment. Where a protocol is run twice on the same segment, we average the two independent passes (k=2).

Table 2: Judges used per experiment.

Experiment	Judges	Notes
Exp 1 (WMT25)	gemini-3-flash-preview (primary); gemini-3.5-flash, GPT-4.1, GPT-4o	k=2 passes, then average
Exp 2–3 (MQM)	GPT-5, gemini-3-flash, GPT-4.1, GPT-4o, Opus 4.7	three FluencyScore variants where available
Grid (§7.1)	gemini-3-flash & GPT-4.1 on WMT25; GPT-5/4.1/4o on MQM	frameworks × judges

Table 3: WMT25 ESA: segment-level Spearman ρ with human ESA (primary mask n=3,526; chrF++ on its own subset). GEMBA-ESA on gemini-3-flash is our reimplementation (§3.4); GEMBA-ESA on GPT-4.1 is official. *On GPT-4.1, official GEMBA exceeds FluencyScore by $\Delta\rho\approx0.045$ pooled; macro-averaged gap ≈0. Matched Gemini protocols are tied.

Backbone	Framework / metric	Spearman ρ	n
gemini-3-flash-preview	FluencyScore	0.613	3526
gemini-3-flash-preview	GEMBA-ESA (ours)	0.606	3526
gemini-3.5-flash	FluencyScore	0.570	3521
GPT-4.1	GEMBA-ESA (official)	0.547*	3526
GPT-4.1	FluencyScore	0.503*	3526
GPT-4o	FluencyScore	0.470	3523
(learned)	MetricX-24-Hybrid-XL	0.518	3526
(learned)	XCOMET-XL	0.391	3526
(learned)	CometKiwi-XL	0.337	3526
(surface)	chrF++ (own subset)	0.281	4200

3.4 Baseline validation and pre-registration

Where an official score exists, our copy of a metric must match it before we trust the copy. For GEMBA-ESA the bar was Spearman 0.85 against official GEMBA-ESA–GPT-4.1 on a stratified 400-segment subset. Our copy reached only 0.784; we never replace an official GEMBA-ESA cell with our copy. Our copy appears only where no official score exists (Gemini backbone), and captions mark those rows.

For GEMBA-MQM, no official scores exist for our MQM years. We treat MQM cohorts as an internal ablation of FluencyScore variants, with one reference-based external anchor (WMT22 GPT-4 release). The decisive WMT25 contrast was pre-registered (mask, statistics, outcome meanings) before the run finished.

4 Experiment 1: WMT25 ESA

Fair subset: n=3,526 segments across the 12 pairs in Section 3.2. Learned baselines include MetricX-24 (Juraska et al., 2024), XCOMET (Guerreiro et al., 2024), and CometKiwi (Rei et al., 2022); the surface baseline is chrF++ (Popović, 2015).

Table 4: Pairwise accuracy with bootstrap 95% CIs (primary mask).

Metric	PA	95% CI	n
FluencyScore / gemini-3-flash	0.722	[0.713, 0.732]	3,526
GEMBA-ESA / gemini-3-flash	0.730	[0.720, 0.739]	3,526
FluencyScore / gemini-3.5-flash	0.709	[0.699, 0.719]	3,521
GEMBA-ESA / GPT-4.1 (official)	0.703	[0.693, 0.713]	3,526
FluencyScore / GPT-4.1	0.679	[0.668, 0.689]	3,526
FluencyScore / GPT-4o	0.669	[0.658, 0.679]	3,523
MetricX-24-Hybrid-XL	0.685	[0.675, 0.695]	3,526
XCOMET-XL	0.638	[0.627, 0.648]	3,526
CometKiwi-XL	0.617	[0.606, 0.629]	3,526
chrF++ (own subset)	0.599	[0.589, 0.609]	4,200

Table 5: Paired bootstrap on PA difference vs FluencyScore / gemini-3-flash.

Baseline	ΔPA	p-value	Sig. (α=0.05)?
GEMBA-ESA / gemini-3-flash	−0.0078	0.018	Yes
GEMBA-ESA / GPT-4.1 (official)	+0.0195	< 0.001	Yes
MetricX-24-Hybrid-XL	+0.0369	< 0.001	Yes
XCOMET-XL	+0.0846	< 0.001	Yes
CometKiwi-XL	+0.1048	< 0.001	Yes
chrF++ (joint subset)	+0.1287	< 0.001	Yes

FluencyScore-Gemini wins 8 of 11 pairs by Spearman ρ in the original per-pair sweep. Losses: EN→mas_KE and EN→ja_JP (GEMBA-ESA leads by ρ) and EN→ar_EG (chrF++ leads by ρ on its subset; FluencyScore still leads by PA).

Against matched GEMBA-ESA on Gemini, the correlation is positive (ρ=+0.62, nominal p=0.03): the structured rubric helps most where the shared judge is weakest (EN→ja_JP is a visible exception). Against official GPT-4.1-backed GEMBA-ESA the sign flips (−0.37, n.s.). With 12 points and six tests, these are exploratory.

5 Experiment 2: WMT22 MQM

Question. Holding the year fixed (WMT22 MQM, n=808, en–de / zh–en / en–ru), do scoring scale (positive 1–10 vs MQM-style negative) and structure (structured rubric vs flat) change agreement when the judge backbone varies?

We separate confounds across three tables. Table 8 is FluencyScore-only; Table 9 would be matched vs GEMBA (cells empty: no API credits / no official GEMBA-MQM); Table 10 lists confounded external anchors only.

Reading (2a1). On GPT-5, structured ≈ flat ≈ MQM-style (0.624 / 0.614 / 0.601). Backbone rank within structured FluencyScore: GPT-5 > Opus 4.7 > Gemini-3-flash > GPT-4o > GPT-4.1.

Table 6: Per-pair Spearman ρ (primary mask). syn_dist: URIEL syntax_knn cosine from English (Littell et al., 2017). GEMBA-ESA = official GPT-4.1; MetricX = MetricX-24.

Pair	n	syn dist	FluencyScore	GEMBA ESA	MetricX
EN→ar_EG	342	0.366	0.415	0.374	−0.303
EN→bho_IN	288	0.337	0.589	0.401	0.300
EN→cs_CZ	304	0.335	0.522	0.401	0.388
EN→et_EE	288	0.195	0.632	0.503	0.326
EN→is_IS	252	0.154	0.797	0.738	0.570
EN→it_IT	252	0.115	0.587	0.435	0.476
EN→ja_JP	288	0.456	0.331	0.377	0.287
EN→mas_KE	306	0.431	0.231	0.389	0.061
EN→ru_RU	288	0.183	0.468	0.353	0.380
EN→sr_Cyrl_RS	306	0.185	0.624	0.543	0.335
EN→uk_UA	342	0.153	0.422	0.379	0.323
EN→zh_CN	270	0.268	0.348	0.344	0.318
MACRO-AVG	3526	—	0.497	0.437	0.288
POOLED	3526	—	0.613	0.547	0.518

Table 7: Typology (exploratory): Spearman correlation between syntactic distance and FluencyScore per-pair PA advantage. Primary mask, 12 pairs; chrF++ on joint subset (11 pairs). Nominal p over six tests.

Baseline	$\rho(\mathrm{syn\_dist}, \mathrm{PA\_adv})$	p-value
GEMBA-ESA / gemini-3-flash	+0.615	0.033
chrF++ (11 pairs)	−0.636	0.035
GEMBA-ESA / GPT-4.1 (official)	−0.371	0.236
CometKiwi-XL	+0.154	0.633
MetricX-24-Hybrid-XL	−0.063	0.846
XCOMET-XL	+0.007	0.983

Takeaways (Experiment 2). On GPT-5, positive structured, MQM-style, and flat variants are close (Table 8). Within FluencyScore, the judge model moves agreement more than scale/structure (GPT-5 vs GPT-4.1: 0.624 vs 0.507). Matched vs GEMBA on WMT22 MQM is unavailable here (Table 9); do not read Tables 10–12 as protocol-matched evidence. Primary matched claim remains WMT25 same-Gemini (Section 7.1).

6 Experiment 3: WMT22–24 MQM

Question. Do the WMT22 patterns from Experiment 2 remain when we expand to three MQM years (extended fair subset: n=2,628 for neural FluencyScore variants)?

Surface baseline. chrF++ recomputed as in Section 3.2: ρ=0.290, PA=0.626, n=2,553. On the WMT22 slice alone the recomputed scores reproduce the previous release figure (ρ≈0.351, PA≈0.653, n=818).

Reading (3a). Flat GPT-5 has the highest ρ on the extended pool (0.584). All FluencyScore rows remain well above recomputed

Table 8: WMT22 MQM (fair n=808): FluencyScore family only. Cells: Spearman ρ / PA. — = not run.

Backbone	Structured (1–10)	MQM-style	Flat
GPT-5	0.624 / 0.780	0.601 / 0.766	0.614 / 0.775
Opus 4.7	0.587 / 0.761	—	—
Gemini-3-flash	0.566 / 0.750	0.586 / 0.755	—
GPT-4o	0.526 / 0.732	—	—
GPT-4.1	0.507 / 0.724	—	—

Table 9: Matched-backbone FluencyScore vs GEMBA (QE). All GEMBA cells are — (no credits / no official GEMBA-MQM). Matched Gemini contrast is on WMT25 (§7.1).

Backbone	FluencyScore (best)	GEMBA (same backbone)
GPT-5	0.624	—
Opus 4.7	0.587	—
Gemini-3-flash	0.566	—
GPT-4o	0.526	—
GPT-4.1	0.507	—

chrF++ (ρ=0.290). Adding WMT23/24 softens the surface baseline relative to the WMT22-only figure (0.351→0.290).

Takeaways (Experiment 3).

- On the extended set the ranking changes relative to Experiment 2 (WMT22-only): flat GPT-5 is above structured GPT-5 in Spearman ρ (0.584 vs 0.551), but the ΔPA is non-significant (Table 14).
- Winners differ by year×pair (Table 15); no single FluencyScore variant is best in every cell.
- Recomputed chrF++ (n=2,553) stays well below the FluencyScore rows (ρ=0.290 vs $\geq$0.50).
- Quote MQM results with the year scope: WMT22-only and WMT22–24 extended are not interchangeable.

7 Experiment 4: Backbone ablation and framework×backbone grid

Experiments 1–3 often change metrics while the judge varies across rows. This section separates the two effects.

Observation. GPT-5 produces a narrower score spread while correlating more strongly with human judgments (interpretation, not a causal claim).

7.1 Framework by backbone grid: primary matched contrasts

This subsection crosses frameworks with backbones on the same segments and the same primary mask as Experiment 1 (n=3,526).

Table 10: External anchors (confounded; orientation only).

Metric	Spearman ρ	PA	Main confounds vs FluencyScore-GPT5
GEMBA-SQM-GPT4-refA	0.515	0.698	GPT-4 $\neq$ GPT-5; has ref
GEMBA-DA-GPT4-refA	0.470	0.673	same
chrF++	0.350	0.653	surface metric

Table 11: Bootstrap 95% CI for Spearman ρ (WMT22 MQM).

Metric	ρ	95% CI	Block
FluencyScore-GPT5	0.624	[0.582, 0.666]	FluencyScore
FluencyScore-MQM-flat-GPT5	0.614	[0.572, 0.663]	FluencyScore
FluencyScore-MQM-GPT5	0.601	[0.557, 0.644]	FluencyScore
FluencyScore-Opus4.7	0.587	[0.541, 0.631]	FluencyScore
FluencyScore-gemini-3.5-flash	0.587	[0.541, 0.632]	FluencyScore
FluencyScore-MQM-Gemini	0.575	[0.525, 0.619]	FluencyScore
FluencyScore-GPT4o	0.526	[0.477, 0.579]	FluencyScore
FluencyScore-GPT4.1	0.507	[0.454, 0.560]	FluencyScore
GEMBA-SQM-GPT4-refA	0.515	[0.463, 0.569]	External
GEMBA-DA-GPT4-refA	0.470	[0.416, 0.529]	External
chrF++	0.350	[0.290, 0.409]	External

In short: the +0.065 headline margin versus official GEMBA-ESA–GPT-4.1 splits into about +0.058 from the judge model and +0.007 from the framework; the framework part has an interval covering zero.

7.2 Variance decomposition

Table 19 attributes variance in Spearman ρ to backbone, framework, and their interaction on three cohorts.

On WMT25 the backbone share stays high. Scale: the positive 1–10 rubric beats the MQM-style negative variant on all four backbones (every interval excludes zero). Structure: the structured variant moves by 0.022 across three shared backbones, against 0.128 for the flat single-score variant.

7.3 The full MQM matrix

Table 20 collects MQM cells from Experiments 2–4 so framework and judge can be read together on the same grid (FluencyScore = positive structured; FluencyScore-MQM = same rubric, negative scale; flat = single-score negative variant; n/a = not run).

8 Error analysis: detects vs misses

Roadmap. §8.1 defines disagreement; §8.2 shows misses; §8.3 shows detects; §8.4 covers MQM variance collapse; §8.5 lists fix directions.

Table 12: Paired bootstrap (PA), WMT22 MQM (confounded comparisons).

Comparison	ΔPA	p-value	Matched?
FluencyScore-GPT5 vs GEMBA-SQM-GPT4-refA	$+0.0818$	0.0035	No
FluencyScore-MQM-Gemini vs GEMBA-SQM-GPT4-refA	$+0.0511$	0.0008	No

Table 13: Segment-level correlation on WMT22–24 MQM (extended). Neural: fair n=2,628. chrF++: source-aligned refA subset n=2,553.

Metric	Spearman ρ	PA	n
FluencyScore-MQM-flat-GPT5	0.584	0.764	2628
FluencyScore-GPT5	0.551	0.746	2628
FluencyScore-Gemini	0.537	0.737	2628
FluencyScore-MQM-Gemini	0.519	0.735	2628
FluencyScore-MQM-GPT5	0.501	0.726	2628
chrF++ (recomputed)	0.290	0.626	2553

8.1 Method

Fair subsets: WMT25 ESA n=3,526; MQM disagreement slice n=824. **Disagreement $|\Delta z|$.** For each segment we z-score FluencyScore and the human score within the analysis corpus, then take $|z_{\text{FS}} - z_{\text{human}}|$. Misses: top-1000 largest-$|\Delta z|$ segments. Detects: top-100 smallest-$|\Delta z|$ segments (deprioritizing ceiling–ceiling).

8.2 What FluencyScore misses (overestimate)

Among the top-1000 largest-disagreement segments, FluencyScore overestimates quality in 46.0% of WMT25 ESA cases and about 39.7% of WMT22–24 MQM cases.

Miss examples. Register miss (EN→ar_EG): human ESA=0.0, FluencyScore=4.26 (MSA instead of Egyptian). None-issue / scale artifact (EN→sr_Cyrl_RS): human ESA=100.0, FluencyScore=8.95 (both like the adaptation; $|\Delta z|$ still large).

Worked judge records. (1) Calque; agree: EN→ru_RU, human ESA 26.5, FluencyScore 1.4 (Minecraft "shulker boxes"). (2) Register; overestimate: EN→ru_RU, human 42.5, minor on *Da, ser*. (3) None; overestimate: EN→it_IT, human 49.0 vs pair median 85.5, FluencyScore 6.1 (ASR cleanup rewarded).

8.3 What FluencyScore detects

On the best-agreement slice FluencyScore is consistently closer to human than GEMBA (100% in the WMT25 ESA top-100). Typical detect: human low, FluencyScore low, `issue_type` correctly identified.

Table 14: Paired bootstrap (PA), extended set.

Comparison	ΔPA	p-value
FluencyScore-GPT5 vs FluencyScore-MQM-flat-GPT5	-0.0188	0.2175 (n.s.)
FluencyScore-GPT5 vs FluencyScore-MQM-Gemini	$+0.0283$	0.0525 (borderline)

Table 15: Per year×pair winners (Spearman ρ). FluencyScore-GPT5 and flat-GPT5 shown for reference.

Year	Pair	Winner	ρ	FluencyScore-GPT5	flat-GPT5
wmt22	en–de	flat-GPT5	0.690	0.545	0.690
wmt22	en–ru	FluencyScore-GPT5	0.674	0.674	0.652
wmt22	zh–en	FluencyScore-GPT5	0.675	0.675	0.519
wmt23	en–de	FluencyScore-MQM-Gemini	0.843	0.738	0.742
wmt23	he–en	flat-GPT5	0.637	0.546	0.637
wmt23	zh–en	FluencyScore-GPT4o	0.624	0.514	0.612
wmt24	en–de	flat-GPT5	0.531	0.401	0.531
wmt24	en–es	flat-GPT5	0.493	0.404	0.493
wmt24	ja–zh	FluencyScore-GPT4.1	0.585	0.497	0.501

8.4 MQM three-judge: variance collapse

8.5 Priority fixes

(1) Calque over-penalization; (2) register/dialect gating; (3) meaning-loss check before high scores (`none`); (4) critical-error detection for Q1; (5) low-confidence flags on low-resource pairs; (6) better false-friend span localization.

8.6 Error-analysis conclusions

FluencyScore is competitive on aggregate correlation, but failure analysis shows structured blind spots. Main overestimate modes: "everything is fluent" (`none`) and register/dialect—not calque. Main underestimate mode: calque over-penalizing acceptable literalism.

9 Limitations

- Judge contamination risk: Gemini training data may overlap with the WMT25 publication window (unquantified).
- No new matched open-weight GEMBA+FluencyScore historical reruns in this revision (budget). Section 7.1 provides matched Gemini contrasts on WMT25.
- Fair-subset dropout on WMT25 is non-uniform (66.3%–90.0% by pair). On WMT25, chrF++ stays on its own subset (n=4,200; missing

Table 16: FluencyScore standard deviation by backbone (1–10 scale).

Backbone	std (1–10)	n non-null
FluencyScore-GPT5	0.898	2,737
FluencyScore-Opus4.7	1.100	2,727
FluencyScore-GPT4.1	1.118	2,757
FluencyScore-GPT4o	1.276	2,750
FluencyScore-Gemini	1.731	2,758

Table 17: OpenRouter extended ablation (n≈2,598).

Label	Spearman ρ	PA	n
FluencyScore-MQM-flat-GPT5	0.584	0.765	2,598
FluencyScore-GPT5	0.553	0.747	2,598
FluencyScore-Gemini	0.538	0.738	2,598
FluencyScore-Opus4.7	0.532	0.736	2,598
FluencyScore-GPT4.1	0.532	0.737	2,598
FluencyScore-GPT4o	0.530	0.737	2,598

for EN→it_IT). For WMT22–24 MQM we recompute chrF++ (Section 3.2, Experiment 6; ρ=0.290, PA=0.626, n=2,553).

- Model snapshot identity: headline Gemini run stored the API alias but not a pinned version string.
- Proprietary prompt/weights limit full public reproduction (Appendix B + confidential review access).
- Error-analysis MQM subset (n=824) differs from correlation MQM subsets (n=808 / 2,628) by mask construction.

10 Discussion and conclusions

We close by restating the claims the data support, the scope boundaries they do not cross, and the design implications for backbone choice.

10.1 Main claims supported by the data

On WMT25 ESA (n=3,526), FluencyScore-Gemini reaches Spearman 0.613, while matched GEMBA-ESA on the same Gemini backbone is statistically tied on ρ and slightly ahead on PA (0.730 vs 0.722). The headline gap vs official GEMBA-ESA–GPT-4.1 is mostly backbone (+0.058 of +0.065); framework CI covers zero. Gaps vs MetricX / XCOMET / CometKiwi / chrF++ by PA remain significant (p<0.05) on the primary mask. On WMT22 MQM, FluencyScore-GPT5 reaches ρ=0.624.

10.2 Scope boundaries

On extended WMT22–24 MQM, flat GPT-5 leads structured GPT-5 in ρ (0.584 vs 0.551; ΔPA n.s.). Recomputed chrF++ on n=2,553 remains far below (ρ=0.290). WMT25 advantages do not transfer uniformly to the full historical MQM cohort. Error analysis: aggregate lead coexists with fluency-bias and register/dialect blind spots.

Table 18: Matched contrasts (primary mask). GEMBA on Gemini is our reimplementation (§3.4).

Contrast	$\Delta\rho$ (pooled)	95% CI	Reading
FluencyScore vs GEMBA-ESA, both gemini-3-flash	+0.007	[−0.009, +0.023]	Matched; CI covers 0
Same, macro-avg (12 pairs)	+0.008	[−0.015, +0.029]	Matched; CI covers 0
GEMBA Gemini (ours) vs official GPT-4.1	+0.058	[+0.038, +0.077]	Mostly backbone
FluencyScore-Gemini vs official GEMBA GPT-4.1	+0.065	[+0.046, +0.084]	Headline
FluencyScore vs GEMBA-ESA, both GPT-4.1	−0.045	[−0.065, −0.024]	Matched GPT-4.1

Table 19: Two-way split of variance in Spearman ρ; bootstrap 95% CIs omitted for space (see full draft).

Cohort	Grid	Backbone	Framework	Interaction
WMT22	3 var.×3 bb (n=815)	97.2%	0.5%	2.3%
WMT22–24	3×3 (n=2,628)	42.5%	31.8%	25.7%
WMT25	2×2 (n=3,526)	87.3%	4.4%	8.3%

10.3 Backbone design

The judge backbone materially affects MQM correlation. Typology: FluencyScore's advantage over chrF++ correlates with syntactic distance; the signal is weaker for neural metrics. We argue for a pinned open-weight judge next.

10.4 Conclusion

FluencyScore is a strong automatic fluency metric on WMT25 ESA under fair masks, competitive on WMT22 MQM, and highly contingent on the judge backbone and evaluation scope. The honest reading of the Gemini-matched grid is that protocol gains are small once the backbone is fixed; the broader WMT22–24 data favor FluencyScore-MQM-flat-GPT5. Error analysis makes behavior legible: calque/false-friend detects work; fluent-but-wrong and dialect/register remain priority fixes. A companion deployment system uses FluencyScore as an online quality signal for Algebras Agentic Localization (https://algebras.ai).

Table 20: Segment-level Spearman ρ with human MQM. †Recomputed chrF++ on extended set (n=2,553).

Backbone	FS (22)	MQM (22)	flat (22)	FS (22–24)	MQM (22–24)	flat (22–24)
gemini-3-flash	0.576	0.591	n/a	0.540	0.519	n/a
GPT-5	0.628	0.603	0.617	0.553	0.502	0.584
GPT-4.1	0.508	0.522	0.508	0.531	0.463	0.453
GPT-4o	0.533	0.518	0.533	0.530	0.486	0.498
Opus 4.7	0.589	n/a	n/a	0.531	n/a	n/a
GEMBA-SQMGPT4refA	0.520	n/a	n/a	n/a	n/a	n/a
chrF++	0.350	n/a	n/a	0.290†	n/a	n/a

Table 21: Issue-type taxonomy (illustrative).

Issue type	Meaning	Example
calque	Source-mirrored structure	EN "make a decision" → RU *sdelat' reshenie*
register_mismatch	Wrong style/dialect/locale	Formal "Good day, sir" in a casual vlog
false_friend	Look-alike wrong meaning	"magazine" → *magazin* (shop)
idiom_literal	Idiom word-by-word	"raining cats and dogs" literal
collocation	Unnatural combination	"heavy rain" → *tyazhyolyy dozhd'*
discourse_marker	Wrong connector/filler	"well," → literal *khorosho,*
none	No fluency issue flagged	n/a

11 Reproducibility Statement

Three layers. (1) **Public:** paper sources, official baseline numbers, masks and n, bootstrap seeds/procedure, scrubbed analysis notebooks, and numeric tables are released at `https://github.com/algebras-ai/fluencyscore-wmt-artifacts` (Appendix A). (2) **API-reproducible:** our GEMBA-ESA cells use published GEMBA-ESA prompts on named public models at temperature 0; provider drift applies; validation in Section 3.4. (3) **Not fully public:** FluencyScore weights and judge prompt. We publish the output schema and taxonomy (Appendix B), release per-segment score tables where safe, include worked examples, state that weights are fixed constants not fitted on evaluation cohorts, and provide weights+prompt to reviewers/area chairs confidentially on request.

A Artifacts and reproduction

- `artifacts/notebooks/fluency2_wmt_evaluation_SAFE.ipynb` — correlation
- `artifacts/notebooks/fluency2_error_analysis_SAFE.ipynb` — error analysis (prompt-edit cells redacted)
- `artifacts/tables/` — precomputed numeric tables

Table 22: Issue-type breakdown of WMT25 ESA top-1000 disagreements.

issue_type	count	% over	mean $\lvert\Delta z\rvert$
calque	467	28.9%	1.27
collocation	97	54.6%	1.40
discourse_marker	11	36.4%	1.22
false_friend	57	22.8%	1.27
idiom_literal	104	41.3%	1.41
none	69	98.6%	3.08
register_mismatch	195	73.8%	1.49

Table 23: Headline failure modes (WMT25 ESA top-1000).

#	Failure mode	WMT25	Direction	Interpretation
1	none	69	∼99% OVER	Fluency bias
2	register_mismatch	195	∼74% OVER	Dialect/locale blindness
3	calque	467	∼29% OVER	Dual mode
4	Q1 collapse	cross	mixed	Worst human quintile
5	low-resource pairs	mas/bho	noisy	Weak correlation

- `artifacts/mqm/` — MQM summary CSVs (segment parquets omitted)
- `artifacts/tables/table3a_wmt2224_with_chrf_full.csv` — recomputed chrF++ aggregates (Exp. 3)

Public release (paper PDF/TEX + safe artifacts): `https://github.com/algebras-ai/fluencyscore-wmt-artifacts`.

Repository layout: repo-root `artifacts/` and `paper/`.

Product site: `https://algebras.ai`. The judge prompt, sub-dimension weights, and deployment routing logic are proprietary and are not included in the public artifacts.

B Output schema and schematic judge prompt

This appendix lists the per-segment fields returned by the judge and a schematic prompt with proprietary content redacted.

B.1 Output schema (per segment)

B.2 Schematic prompt (placeholders only)

System role. You are a translation fluency judge. Score only fluency-related phenomena; do not optimize for adequacy alone.

Input. `source_lang`, `target_lang`, `source_text`, `mt_text`; optional locale / formality / glossary.

Rubric `[PROPRIETARY_RUBRIC_TEXT]`. For each of idiomatic, collocational, discourse,

Table 24: Agreement-slice summary statistics.

Metric	Value
WMT25 fair n	3274
Mean $\|\Delta z\|$ FluencyScore (all fair)	0.777
Mean $\|\Delta z\|$ GEMBA (all fair)	0.650
Top-100 agreement mean $\|\Delta z\|$	0.024
% rows FluencyScore closer than GEMBA (fair)	43.2%
Top-100: FluencyScore closer than GEMBA	100%
MQM top-100 agreement mean $\|\Delta z\|$	0.058

Table 25: FluencyScore-MQM (Gemini): 41.6% of mass at ≥98—ceiling collapse.

mode / judge	n	% ≥98	std (0–100)
FluencyScore / Opus4.7	2727	0.6%	18.32
FluencyScore / GPT5	2737	0.4%	17.03
FluencyScore-MQM / Gemini	2747	41.6%	9.66
FluencyScore-MQM / GPT5	2758	23.0%	15.91

pragmatic, calque → score in $[1, 10]$.

Aggregation `[PROPRIETARY_WEIGHTS]`: aggregate $= \sum_i w_i \cdot \mathrm{dim}_i$.

Error record: choose one `issue_type`; set severity; quote `problematic_phrase`; propose `suggested_fix`; set confidence fields.

Output: JSON matching the schema above only. No chain-of-thought in the returned object. *Decoding:* temperature 0; two independent calls; mean aggregates.

B.3 Confidential appendix

Full prompt text and numeric weights are available to reviewers and area chairs on request under the venue's confidential-review process.

Table 26: Per-segment output schema.

Field	Type	Meaning
idiomatic	float 1–10	Idiomatic phrasing
collocational	float 1–10	Collocational fit
discourse	float 1–10	Discourse / connectors
pragmatic	float 1–10	Pragmatic appropriateness
calque	float 1–10	Freedom from calque (higher=better)
aggregate	float	Fixed weighted sum of the five dims
issue_type	enum	Main issue label
severity	enum	cosmetic|minor|major|critical
problematic_phrase	string	Span in the MT
suggested_fix	string	Suggested rewrite
low_confidence	bool	Judge uncertainty flag
max_score_diff	float	Disagreement across double-pass

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